

Introduction to Microeconomics

Consumer Theory II: The Consumer's Optimization Problem & Demand systems

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The Consumer's Problem

The Consumer's Problem

Choose the best bundle, according to the consumer's preferences subject to the consumer's budget constraint. → Choose the best among what is feasible.

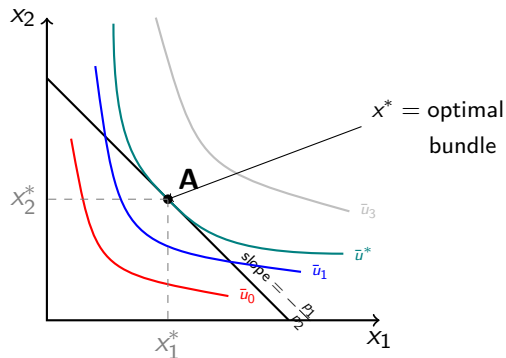
$$\begin{aligned} \max_{(x_1, x_2, \dots, x_n) \in \mathbb{R}_+^n} & \quad u(x_1, x_2, \dots, x_n) \\ \text{subject to:} & \quad p_1 x_1 + p_2 x_2 + \dots + p_n x_n \leq I \end{aligned}$$

where $p_1, p_2, \dots, p_n$ are the prices of the n goods and I is income.

Two-good world

$$\begin{aligned} \max_{(x_1, x_2) \in \mathbb{R}_+^2} & \quad u(x_1, x_2) \\ \text{subject to:} & \quad p_1 x_1 + p_2 x_2 \leq I \end{aligned}$$

Graphical Analysis of the Consumer's Problem



Point A is the optimal consumption bundle!

$$1. -\frac{p_1}{p_2} = MRS \iff |MRS| = \frac{p_1}{p_2}$$

$$2. p_1x_1 + p_2x_2 = I$$

(exhausting the budget)

$\hookrightarrow$ at an optimum (1) & (2) hold.

BL: $p_1x_1 + p_2x_2 = I$

Budget set: $p_1x_1 + p_2x_2 \leq I$

Preferences: Indifference curve map of the consumer

$$(1) \iff \frac{MU_1}{MU_2} = \frac{p_1}{p_2} \iff \frac{MU_1}{p_1} = \frac{MU_2}{p_2}$$

$$\max_{(x_1, x_2) \in \mathbb{R}_+^n} u(x_1, x_2) \quad \text{subject to: } p_1 x_1 + p_2 x_2 \leq I$$

Suppose that preferences are:

- ① Complete
- ② Transitive
- ③ Monotonic (More is better)
- ④ Strictly Convex (strict preference for balancedness) $\leftrightarrow$ (MRS strictly decreasing)

The optimality condition

We can simplify the problem to: $\max_{(x_1, x_2) \in \mathbb{R}_+^n} u(x_1, x_2)$ subject to: $p_1 x_1 + p_2 x_2 = I$
Furthermore, the consumer's optimal consumption bundle satisfies:

$$MRS = \frac{MU_1}{MU_2} = \frac{p_1}{p_2}$$

If preferences are monotonic the B.C. holds with equality $\implies$

$$\max_{(x_1, x_2) \in \mathbb{R}_+^2} U(x_1, x_2) \quad \text{s.t.} \quad p_1 x_1 + p_2 x_2 = I$$

$$\implies x_2 = I/p_2 - \frac{p_1}{p_2} x_1$$

$$\implies \max_{x_1 \in \mathbb{R}_+} U\left(x_1, I/p_2 - \frac{p_1}{p_2} x_1\right)$$

First Order Condition (FOC):

$$\frac{dU}{dx_1} = \frac{\partial U}{\partial x_1} + \frac{\partial U}{\partial x_2} \cdot \left(-\frac{p_1}{p_2}\right) = 0 \implies \frac{MU_1}{MU_2} = \frac{p_1}{p_2}$$

Cobb-Douglas Utility: $u(x_1, x_2) = x_1 x_2$

Notice that we can apply a monotonic transformation: $v(x_1, x_2) = \ln(u(x_1, x_2)) = \ln x_1 + \ln x_2$.
With this transformation: it is easy to obtain the marginal utilities.

$$p_1 = 1, \quad MU_1 = 1/x_1$$

$$p_2 = 2, \quad MU_2 = 1/x_2$$

$$I = 80$$

$$\Rightarrow |MRS| = \frac{MU_1}{MU_2} = \frac{1/x_1}{1/x_2} = \frac{x_2}{x_1}$$

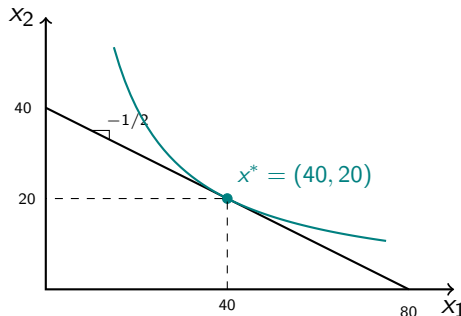
$$\max_{(x_1, x_2) \in \mathbb{R}_+^2} x_1 x_2 \quad \text{s.t.} \quad x_1 + 2x_2 = 80 \Rightarrow x_2 = 40 - \frac{1}{2}x_1$$

$$1. \quad |MRS| = \frac{p_1}{p_2} \iff \frac{x_2}{x_1} = \frac{1}{2} \Rightarrow x_2 = \frac{1}{2}x_1$$

$$2. \quad x_1 + 2x_2 = 80$$

$$x_1 + 2\left(\frac{1}{2}x_1\right) = 80 \Rightarrow$$

$$\boxed{\begin{array}{l} x_1^* = 40 \\ x_2^* = 20 \end{array}}$$



Another Example: $u(x_1, x_2) = \sqrt{x_1} + \sqrt{x_2}$

Parameters:

$$p_1 = 1, \quad p_2 = 2, \quad I = 60$$

$$MU_1 = \frac{1}{2\sqrt{x_1}}, \quad MU_2 = \frac{1}{2\sqrt{x_2}}$$

$$\Rightarrow |MRS| = \frac{\sqrt{x_2}}{\sqrt{x_1}}$$

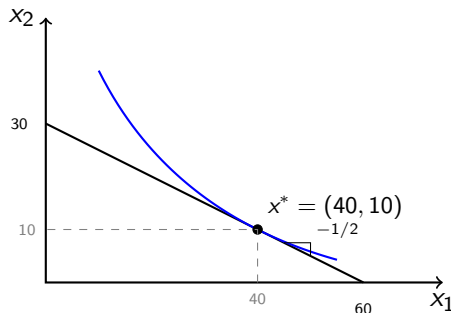
$$\begin{aligned} 1. \quad |MRS| &= \frac{p_1}{p_2} \iff \frac{\sqrt{x_2}}{\sqrt{x_1}} = \frac{1}{2} \\ &\Rightarrow \frac{x_2}{x_1} = \frac{1}{4} \Rightarrow x_1 = 4x_2 \end{aligned}$$

$$2. \quad x_1 + 2x_2 = 60$$

Substitute (1) into (2):

$$4x_2 + 2x_2 = 60 \Rightarrow 6x_2 = 60$$

$x_2^* = 10$
$x_1^* = 40$



Perfect Complements: Utility Maximization

Problem: Suppose I like exactly 2 slices of Bread (B) for every 1 slice of Ham (H) in my sandwiches.

1. Preferences & Constraints

- **Utility:** $U(H, B) = \min\{H, \frac{B}{2}\}$
- **Kink:** $H = \frac{B}{2} \implies B = 2H$
- **Prices:** $P_H = 3, P_B = 1$
- **Income:** $I = 150$

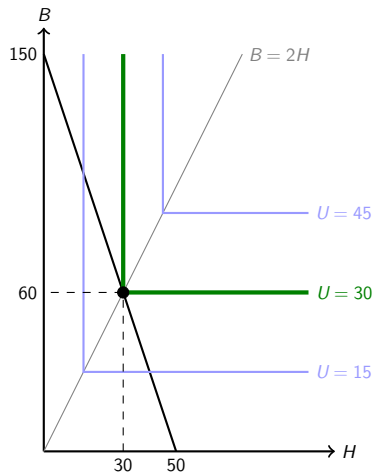
2. Optimal Solution

Substitute $B = 2H$ into $3H + B = 150$:

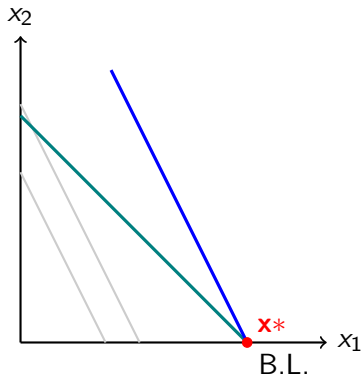
$$3H + (2H) = 150$$

$$5H = 150 \implies H^* = 30$$

$$B = 2(30) \implies B^* = 60$$



Corner Solutions: Perfect Substitutes



Corner Solution intuition

The consumer pushes to the highest possible indifference curve until it hits the boundary of the budget set.

1. Preferences & Constraints

- **Utility:**

$$u(x_1, x_2) = 2x_1 + x_2 \implies |MU_1/MU_2| = |MRS| = 2$$

- **Prices:** $p_1 = 150$, $p_2 = 150$

- **Income:** $I = 500$

- **Price Ratio:** $\frac{p_1}{p_2} = \frac{150}{150} = 1$

2. Optimal Solution

$$|MRS| > \frac{p_1}{p_2} \implies \frac{MU_1}{p_1} > \frac{MU_2}{p_2} \implies \frac{2}{150} > \frac{1}{150}$$

Thus, the consumer chooses a **corner solution** by spending all income on x_1 :

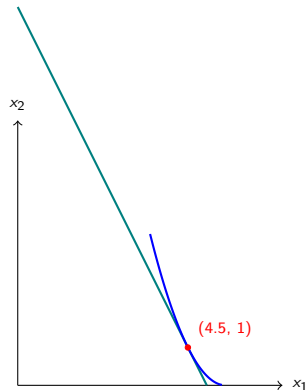
$$x_1^* = \frac{I}{p_1} = \frac{500}{150} \implies x_1^* = 3.33, \quad x_2^* = 0$$

Quasi-linear Case: $u(x_1, x_2) = x_1 + \sqrt{x_2}$

Example 1: Interior Solution (Tangency)

Let $p_1 = 2$, $p_2 = 1$, $I = 10$.

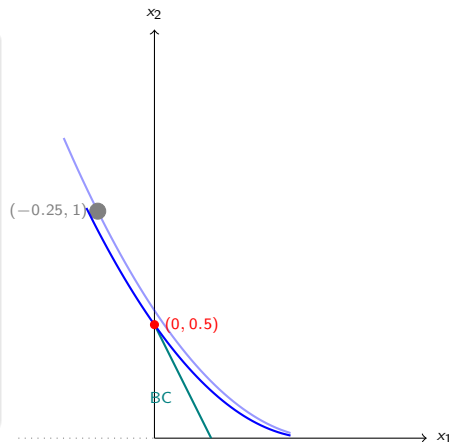
- 1 Set $|MRS| = \frac{p_1}{p_2} \implies 2\sqrt{x_2} = 2 \implies \mathbf{x_2^* = 1}$.
- 2 Check affordability: Cost is $p_2 x_2 = 1$. Since $I > 1$, it is affordable.
- 3 Remaining income for x_1 : $x_1^* = \frac{10-1}{2} = \mathbf{4.5}$.
- 4 **Result:** At $(4.5, 1)$, $\frac{MU_1}{p_1} = \frac{MU_2}{p_2} = 0.5$.



Example 2: Corner Solution

Let $p_1 = 2$, $p_2 = 1$, $I = 0.5$.

- 1 Tangency requires $x_2 = 1$, which implies $x_1 = -0.25$.
- 2 **Theoretical** IC_{tan} : Passes through $(-0.25, 1)$ with $U = 0.75$.
- 3 **Actual** IC_{corner} : Passes through $(0, 0.5)$ with $U = \sqrt{0.5} \approx 0.707$.
- 4 **Result**: Since $x_1 \geq 0$, the corner $(0, 0.5)$ is the highest reachable utility.



- Recall the problem of the consumer:

$$\max_{x_1, x_2} u(x_1, x_2) \text{ subject to } p_1 x_1 + p_2 x_2 \leq I,$$

where p_1 and p_2 are strictly positive prices and I is a strictly positive level of income.

- The solution to the problem above are the optimal levels of consumption of goods 1 and 2, when the prices are p_1 and p_2 and income is I .
- When prices and income are left as unspecified, **the optimal levels of consumption are functions of those parameters.**
- The optimal solution is written as: $x_1^m(p_1, p_2, I)$ and $x_2^m(p_1, p_2, I)$. These functions are called **the Marshallian demand system of the consumer.**

Procedure to obtain the demand system from the UMP

The basic steps are:

- 1 Obtain the marginal utilities for good 1 and good 2. (Usually given as data for the problem.)
- 2 Obtain the marginal rate of substitution, where you need to decide which good goes in the horizontal axis and which good goes in the vertical axis.
- 3 Write the optimality condition for an interior optimum. (In the case you have a problem with an interior optimum and not a corner solution.)
- 4 From the optimality condition, solve for one of the variables.
- 5 Substitute what you obtained in the previous step into the budget constraint and solve for the optimal level of one of the goods as a function of the parameters (prices and income). This will be the Marshallian demand function for one of the goods.
- 6 With the Marshallian demand of one of the goods, use the expression you obtained in step 4 to solve for the other Marshallian demand. The two expressions you obtained now form the **Marshallian Demand System** for the consumer.

Cobb-Douglas Utility

$$u(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$$

where $0 < \alpha < 1$.

The marginal utilities are: $MU_1 = \alpha x_1^{\alpha-1} x_2^{1-\alpha}$ and $MU_2 = (1-\alpha) x_1^\alpha x_2^{-\alpha}$. Then, the marginal rate of substitution is

$$MRS = \frac{MU_1}{MU_2} = \frac{\alpha x_1^{\alpha-1} x_2^{1-\alpha}}{(1-\alpha) x_1^\alpha x_2^{-\alpha}} = \frac{\alpha x_2}{(1-\alpha) x_1}.$$

The optimality condition for an interior optimum is:

$$\frac{\alpha x_2}{(1-\alpha) x_1} = \frac{p_1}{p_2} \Rightarrow x_2 = \frac{1-\alpha}{\alpha} \frac{p_1}{p_2} x_1.$$

Substituting the previous expression into the budget constraint yields:

$$p_1 x_1^m + p_2 \left(\frac{1-\alpha}{\alpha} \frac{p_1}{p_2} x_1^m \right) = I \Rightarrow p_1 x_1^m \left(1 + \frac{1-\alpha}{\alpha} \right) = I \Rightarrow p_1 x_1^m \left(\frac{1}{\alpha} \right) = I \Rightarrow x_1^m = \frac{\alpha I}{p_1}.$$

Cobb-Douglas Utility

Using the results from the previous two displayed equations yields:

$$x_2^m = \frac{1 - \alpha}{\alpha} \frac{p_1}{p_2} \frac{\alpha I}{p_1} = \frac{(1 - \alpha)I}{p_2}$$

The Marshallian Demand System for the Cobb-Douglas utility function is then:

$$x_1^m(p_1, p_2, I) = \frac{\alpha I}{p_1}, \quad x_2^m(p_1, p_2, I) = \frac{(1 - \alpha)I}{p_2}$$

Note that if we define the **expenditure share** of each good as:

$$s_1 = \frac{p_1 x_1}{I}, \quad s_2 = \frac{p_2 x_2}{I},$$

then the expenditure shares for the Cobb-Douglas Marshallian demand system are simply: $s_1 = \alpha$ and $s_2 = 1 - \alpha$.

Constant-Elasticity of Substitution (CES) Utility

$$u(x_1, x_2) = \frac{x_1^\delta}{\delta} + \frac{x_2^\delta}{\delta}$$

where $\delta \leq 1$ and $\delta \neq 0$.

The marginal utilities are: $MU_1 = x_1^{\delta-1}$ and $MU_2 = x_2^{\delta-1}$. Then, the marginal rate of substitution is:

$$MRS = \frac{MU_1}{MU_2} = \frac{x_1^{\delta-1}}{x_2^{\delta-1}} = \left(\frac{x_1}{x_2} \right)^{\delta-1}.$$

The optimality condition for an interior optimum is:

$$\left(\frac{x_1}{x_2} \right)^{\delta-1} = \frac{p_1}{p_2} \Rightarrow \frac{x_1}{x_2} = \left(\frac{p_1}{p_2} \right)^{\frac{1}{\delta-1}}.$$

Constant-Elasticity of Substitution (CES) Utility

You can solve for either x_1 or x_2 and plug into the budget constraint to obtain the demands, but this requires some tedious algebra. Thus, I will show you here another a trick that you can use that also has a lot of economic meaning. Multiply both sides of the previous equation by the price ratio p_1/p_2 to get:

$$\frac{p_1 x_1}{p_2 x_2} = \frac{p_1}{p_2} \left(\frac{p_1}{p_2} \right)^{\frac{1}{\delta-1}} = \left(\frac{p_1}{p_2} \right)^{1+\frac{1}{\delta-1}} = \left(\frac{p_1}{p_2} \right)^{\frac{\delta}{\delta-1}}.$$

The previous equation implies that the ratio of expenditure shares $s_1/s_2 = \left(\frac{p_1 x_1 / I}{p_2 x_2 / I} \right)$ is equal to:

$$\frac{p_1 x_1}{p_2 x_2} = \frac{I p_1 x_1}{I p_2 x_2} = \frac{s_1}{s_2} = \left(\frac{p_1}{p_2} \right)^{\frac{\delta}{\delta-1}},$$

and therefore:

$$s_1 = \left(\frac{p_1}{p_2} \right)^{\frac{\delta}{\delta-1}} s_2.$$

Constant-Elasticity of Substitution (CES) Utility

Now, the budget line in terms of expenditure shares is: $s_1 + s_2 = 1$. We can use the previous two equations to solve for the optimal expenditure shares:

$$\left(\frac{p_1}{p_2}\right)^{\frac{\delta}{\delta-1}} s_2 + s_2 = 1 \Rightarrow s_2 \left(1 + \left(\frac{p_1}{p_2}\right)^{\frac{\delta}{\delta-1}}\right) = 1 \Rightarrow s_2 = \left(1 + \left(\frac{p_1}{p_2}\right)^{\frac{\delta}{\delta-1}}\right)^{-1}$$

Plugging the previous expression into the equation relating the two expenditure shares yields, after some algebra:

$$s_1 = \left(1 + \left(\frac{p_2}{p_1}\right)^{\frac{\delta}{\delta-1}}\right)^{-1}.$$

Notice that the optimal expenditure shares for each good from the CES utility function depend only on the price ratio and the parameter δ . In contrast with the Cobb-Douglas utility, these budget shares do change with changes in the price ratio. However, the expressions remain relatively simple. With a little bit more work (which you may learn in future courses), these shares can be put into a form that can be used for empirical work. The CES specification is widely used in trade theory to analyze the effects of trade tariffs and quotas, since they change the relative price ratios of goods.

Constant-Elasticity of Substitution (CES) Utility

With the optimal expenditure shares, we can compute the Marshallian demands as follows:

$$p_1 x_1^m / I = s_1(p_1/p_2) \Rightarrow x_1^m = \frac{I}{p_1} s_1(p_1/p_2), \quad p_2 x_2^m / I = s_2(p_1/p_2) \Rightarrow x_2^m = \frac{I}{p_2} s_2(p_1/p_2).$$

The Marshallian demand system for the CES utility function is then:

$$x_1^m(p_1, p_2, I) = \frac{I}{p_1} s_1(p_1/p_2), \quad x_2^m(p_1, p_2, I) = \frac{I}{p_2} s_2(p_1/p_2)$$

Perfect Complements Utility

$$u(x_1, x_2) = \min \left(\frac{x_1}{\alpha}, \frac{x_2}{\beta} \right),$$

where $\alpha > 0$ and $\beta > 0$. Marginal utilities and the marginal rate of substitution are undefined for this utility function. However, we know that the indifference curve look like “right-angles”. The set of points that are the kinks in the indifference curve map is given by

$$\frac{x_1}{\alpha} = \frac{x_2}{\beta} \Rightarrow x_1 = \frac{\alpha}{\beta} x_2.$$

Substitute the previous expression into the budget constraint to obtain:

$$p_1 \left(\frac{\alpha}{\beta} x_2^m \right) + p_2 x_2^m = I \Rightarrow x_2^m \left(\frac{\alpha p_1 + \beta p_2}{\beta} \right) = I \Rightarrow x_2^m = \frac{\beta I}{\alpha p_1 + \beta p_2}.$$

Perfect Complements Utility

Using the results from the two previous displayed equations yields:

$$x_1^m = \frac{\alpha}{\beta} \frac{\beta I}{\alpha p_1 + \beta p_2} = \frac{\alpha I}{\alpha p_1 + \beta p_2}.$$

The Marshallian Demand System for the Perfect Complements utility function is then:

$$x_1^m(p_1, p_2, I) = \frac{\alpha I}{\alpha p_1 + \beta p_2}, \quad x_2^m(p_1, p_2, I) = \frac{\beta I}{\alpha p_1 + \beta p_2}$$

Perfect Substitutes Utility

$$u(x_1, x_2) = \alpha x_1 + \beta x_2,$$

where $\alpha > 0$ and $\beta > 0$. The marginal utilities are $MU_1 = \alpha$ and $MU_2 = \beta$. Then, the marginal rate of substitution is

$$MRS = \frac{MU_1}{MU_2} = \frac{\alpha}{\beta}.$$

With perfect substitutes utility we will have corner solutions, so you should not start by equating the price ratio to the marginal rate of substitution.

Thus, if $\frac{\alpha}{p_1} > \frac{\beta}{p_2}$, which is equivalent to $MRS = \frac{\alpha}{\beta} > \frac{p_1}{p_2}$, then all the budget should be used to consume good 1: $x_1^m = \frac{I}{p_1}$, $x_2^m = 0$. On the other hand, if $\frac{\alpha}{p_1} < \frac{\beta}{p_2}$, which is equivalent to $MRS = \frac{\alpha}{\beta} < \frac{p_1}{p_2}$, then all the budget should be used to consume good 2: $x_1^m = 0$, $x_2^m = \frac{I}{p_2}$. If it is the case that $\frac{\alpha}{p_1} = \frac{\beta}{p_2}$, which is equivalent to $MRS = \frac{\alpha}{\beta} = \frac{p_1}{p_2}$, then any point in the budget line is optimal: $x_1^m \in [0, I/p_1]$ and $x_2^m = I/p_2 - (p_1/p_2)x_1^m$.

Perfect Substitutes Utility

Then the Marshallian demand system for the Perfect Substitutes utility function is:

$$x_1^m(p_1, p_2, I) = \begin{cases} \frac{I}{p_1} & \text{if } \frac{\alpha}{\beta} > \frac{p_1}{p_2} \\ 0 & \text{if } \frac{\alpha}{\beta} < \frac{p_1}{p_2} \\ [0, I/p_1] & \text{if } \frac{\alpha}{\beta} = \frac{p_1}{p_2} \end{cases}, \quad x_2^m(p_1, p_2, I) = \begin{cases} 0 & \text{if } \frac{\alpha}{\beta} > \frac{p_1}{p_2} \\ \frac{I}{p_2} & \text{if } \frac{\alpha}{\beta} < \frac{p_1}{p_2} \\ \frac{I}{p_2} - \left(\frac{p_1}{p_2}\right) x_1^m(p_1, p_2, I) & \text{if } \frac{\alpha}{\beta} = \frac{p_1}{p_2} \end{cases}$$

Comparative Statics or Ceteris Paribus (everything else held equal) analysis

We will perform now a comparative statics analysis by holding some parameters constant and changing one parameter at a time.

Individual demand curves

Suppose that income is held constant at a level $\tilde{I}$.

- If the price of good 2 is held constant, say at $\tilde{p}_2$, and we let p_1 vary freely, we obtain the **individual demand function for good 1**: $x_1(p_1) = x_1^m(p_1, \tilde{p}_2, \tilde{I})$.
- Similarly, if we hold the price of good 1 constant, say at $\tilde{p}_1$, and we let p_2 vary freely, we obtain the **individual demand function for good 2**: $x_2(p_2) = x_2^m(\tilde{p}_1, p_2, \tilde{I})$.

We usually plot them in a price vs quantity diagram.

Price Consumption Curve

Suppose that income is held constant at a level $\tilde{I}$ and the price of good 2 is held constant, say at $\tilde{p}_2$, and we allow p_1 to vary freely. The Price Consumption Curve (PCC) with respect to p_1 is the set of optimal choices of good 1 and good 2 as we vary the price of good 1. We plot this curve in consumption space, i.e., good 1 vs good 2. Formally, the PCC is the vector-valued function

$$PCC: \mathbb{R}_+ \rightarrow \mathbb{R}_+^2$$

which is given by

$$PCC(p_1) = (x_1^m(p_1, \tilde{p}_2, \tilde{I}), x_2^m(p_1, \tilde{p}_2, \tilde{I}))$$

Cournot Aggregation

Recall that the price-elasticity of demand of good i with respect to the price of good j is

$$\epsilon_{i,j} = \frac{\Delta\%x_i}{\Delta\%p_j}.$$

Also recall that the share of income spent on good i is

$$s_i = \frac{p_i x_i}{I}.$$

Cournot Aggregation

We have that the following equations hold true for all Marshallian Demand Systems, called the **Cournot Aggregations** for good 1 and good 2 respectively

$$s_1 \epsilon_{1,1} + s_2 \epsilon_{2,1} = -s_1$$

$$s_1 \epsilon_{1,2} + s_2 \epsilon_{2,2} = -s_2$$

We will now hold prices fixed and see what happens when income changes.

Engel Curves

Suppose that both prices are held constant, say at $\tilde{p}_1$ and $\tilde{p}_2$. If we allow income I to vary freely, then for each good we get its **Engel Curve**:

$$E_1(I) = x_1^m(\tilde{p}_1, \tilde{p}_2, I) \quad E_2(I) = x_2^m(\tilde{p}_1, \tilde{p}_2, I).$$

These curves are plotted in Income vs consumption space.

Income-Consumption Curve

Suppose that both prices are held constant, say at $\tilde{p}_1$ and $\tilde{p}_2$. The **Income Consumption Curve** (ICC) is the set of optimal choices of good 1 and good 2 as we vary the level of income. Formally, the ICC is the vector-valued function

$$ICC : \mathbb{R}_+ \rightarrow \mathbb{R}_+^2$$

which is given by

$$ICC(I) = (x_1^m(\tilde{p}_1, \tilde{p}_2, I), x_2^m(\tilde{p}_1, \tilde{p}_2, I))$$

The ICC is plotted in consumption space: good 1 vs good 2.

When the consumer exhausts her budget $p_1x_1 + p_2x_2 = I$, if we divide the whole equation by I we get:

$$s_1 + s_2 = 1.$$

The Engel Aggregation

This is a relationship about income elasticities across a variety of goods. We have that the following equation, called the **Engel Aggregation**, is true:

$$s_1\epsilon_{1,I} + s_2\epsilon_{2,I} = 1$$